

Cryptanalysis Attack

Estimated duration: 5 minutes

Introduction

This lab introduces you to cryptanalysis using the CrypTool 2.1 software. You will explore the process of breaking Data Encryption Standard (DES) encryption through known plaintext/Ciphertext Only attack. By following step-by-step instructions, you will observe how cryptanalysis techniques are used to derive the deciphered text and the encryption key, understanding the underlying vulnerabilities of symmetric encryption algorithms like DES.

Learning objectives

After completing this lab, you will be able to:

- Perform a known plaintext/Ciphertext Only attack on the DES encryption algorithm
- Analyze how to recover a deciphered message and key using cryptanalysis techniques
- Observe the vulnerabilities in symmetric encryption methods like DES

Part 1: Installing CrypTool 2.1

Note: If you have CrypTools already installed, you can skip to Part 2: DES Cryptanalysis.

Step 1: Download CrypTool 2.1

- **CrypTool 2.1** is the current version available as a desktop application.
- Go to the official CrypTool website: <https://www.cryptool.org/en>.
(*Note: To open the links, right-click (or long-press) on the links and select "Open in new tab." Avoid clicking the link directly, as this might block it.*)

Step 2: Install CrypTool 2.1 on Windows

- Download **CrypTool 2.1 (Stable Build 9778.2)** from the official website:
 - Go to [CrypTool 2 Download Page](#).*
 - (*Note: To open the links, right-click (or long-press) on the links and select "Open in new tab." Avoid clicking the link directly, as this might block it.*)
 - Select the version compatible with your operating system (typically a .exe file for Windows).
- Run the installer by double-clicking the downloaded file.
 - Follow the prompts and accept the default installation options.

Part 2: Performing DES Cryptanalysis

- Open **CrypTool 2.1**.

- Select **New** option.

CrypTool 2.1 (Stable Build 9778.2) – Startcenter

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File Execute View

★ Startcenter



CRYPTtOOL 2

Welcome to CrypTool 2. There are two ways for a quick start: Click on the wizard button in the section "Main Functions" to get a guided tour, or load one of the pre-defined workflows in the section

★ Main functions

 Create a new workspace with the graphical editor

 Use the wizard to easily try some CrypTool 2 features

📘 External resources

 View a tutorial video about CrypTool 2 on YouTube

 Visit the CrypTool Book page on the CrypTool portal

 Visit the official CrypTool 2 website

 Visit the official YouTube channel: You will learn how to use CrypTool 2 and how to add your own functions

 Visit us on Facebook

📁 Templates (double click to open)

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▶  Hash Functions

▶  Mathematics

▶  Codes

▶  Protocols

▶  Steganography

▶  Tools

📁 Recently opened workspaces (double click to open)

 Factorization with Quadratic Sieve (QS)

 DES DES Analysis Using Entropy

 RSA RSA Decryption

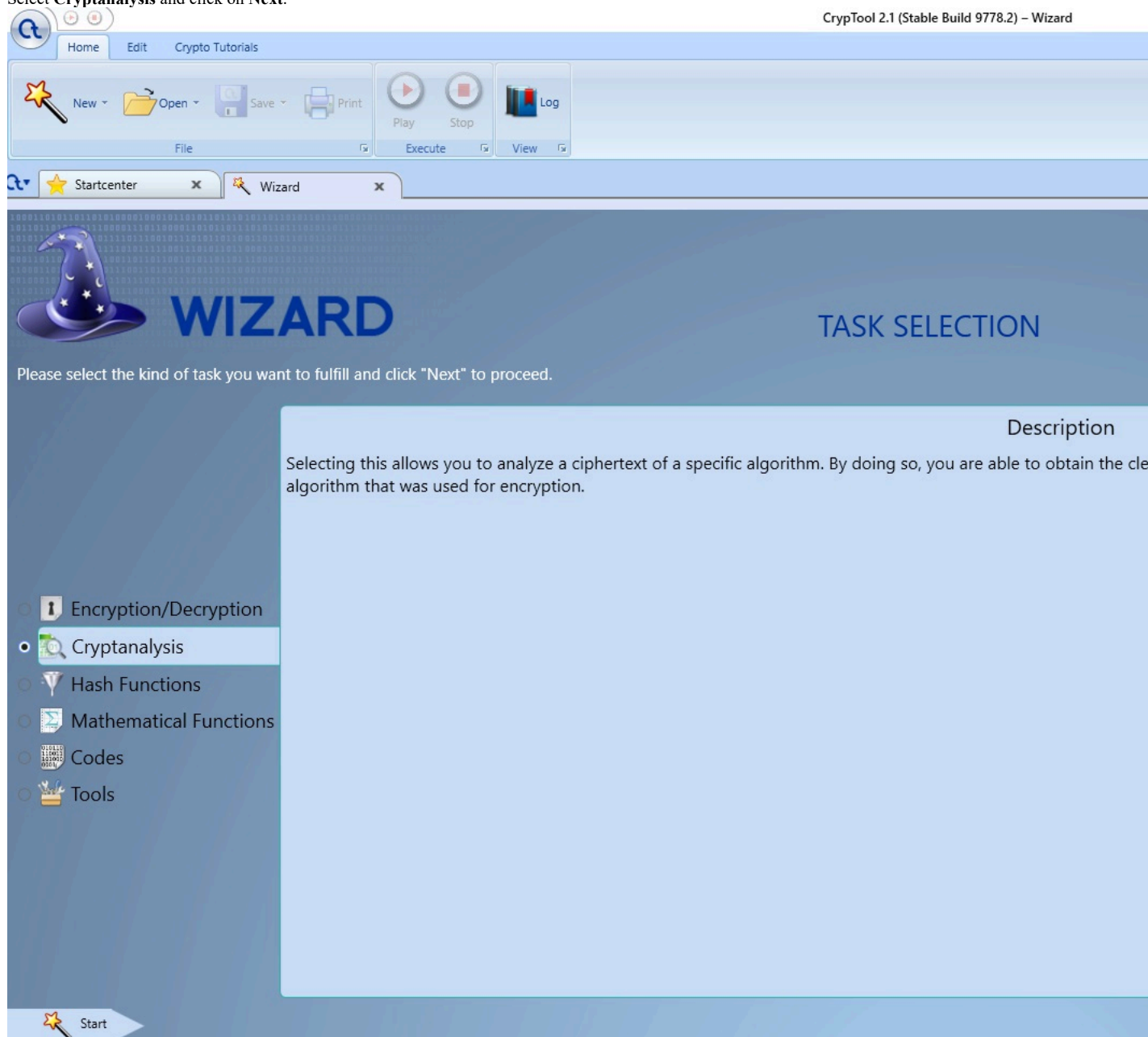
 RSA RSA Encryption

 RSA Key Generator

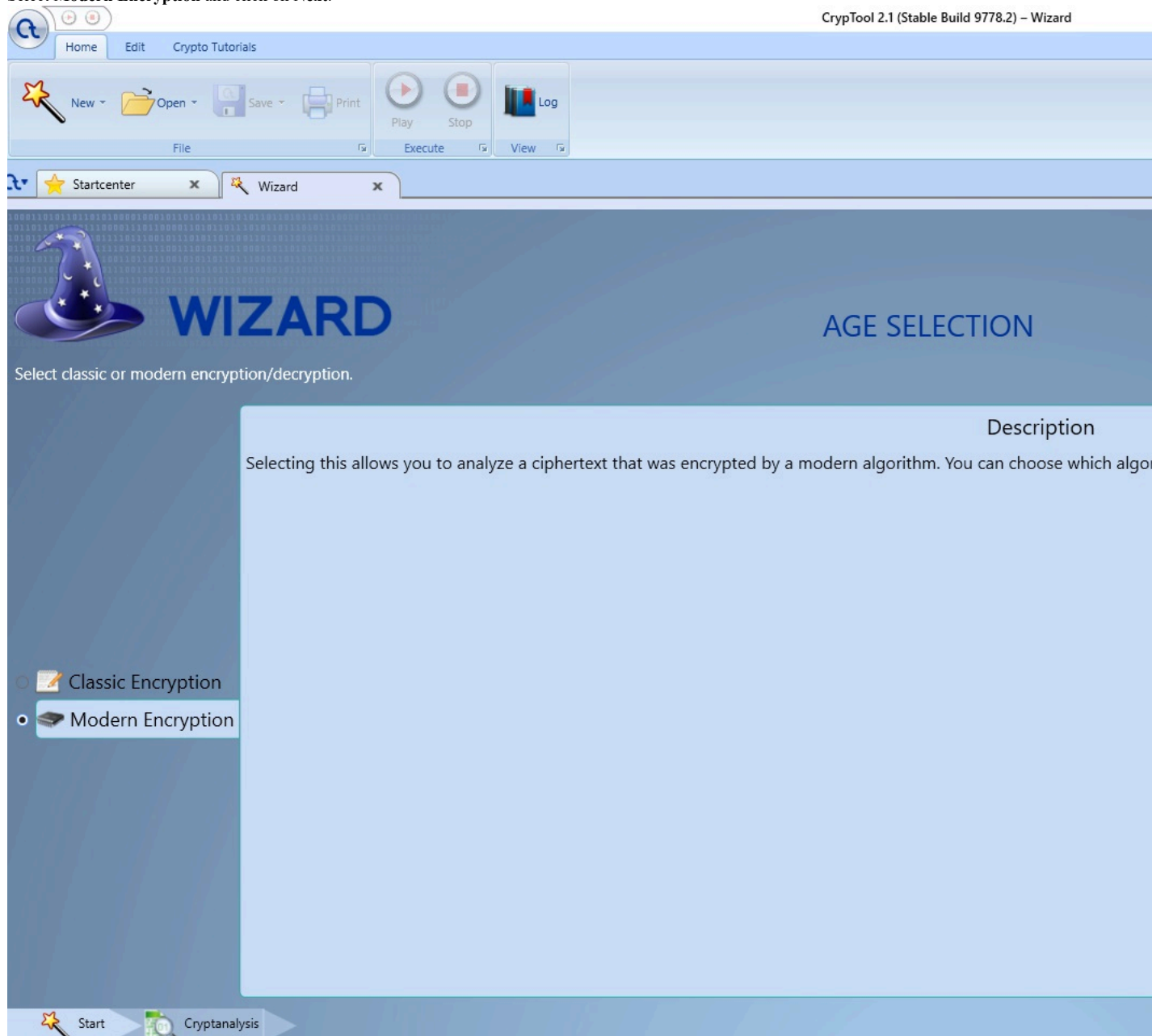
 AES (CBC) with PKCS#5 and IV

 DES

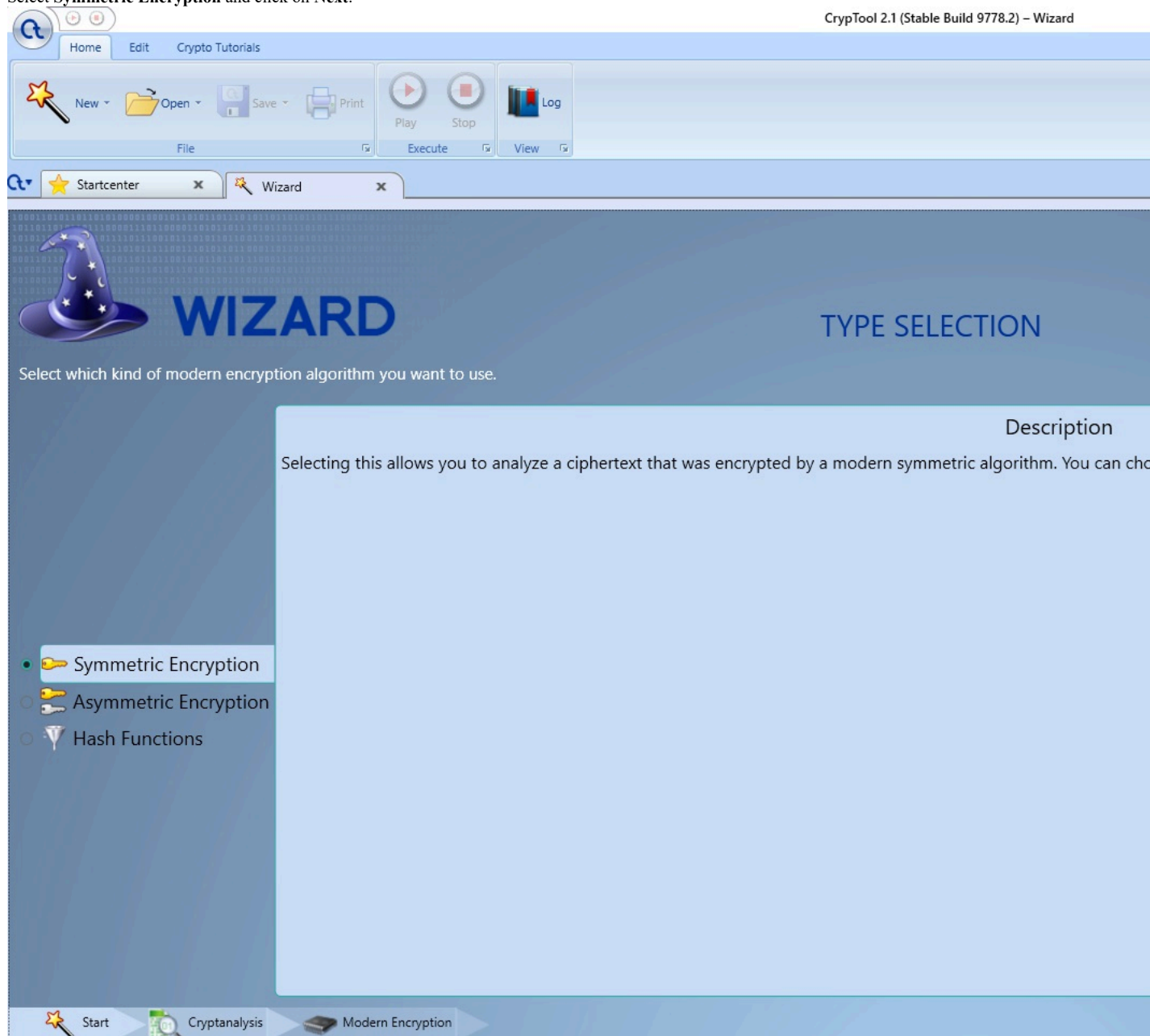
- Select **Cryptanalysis** and click on **Next**.



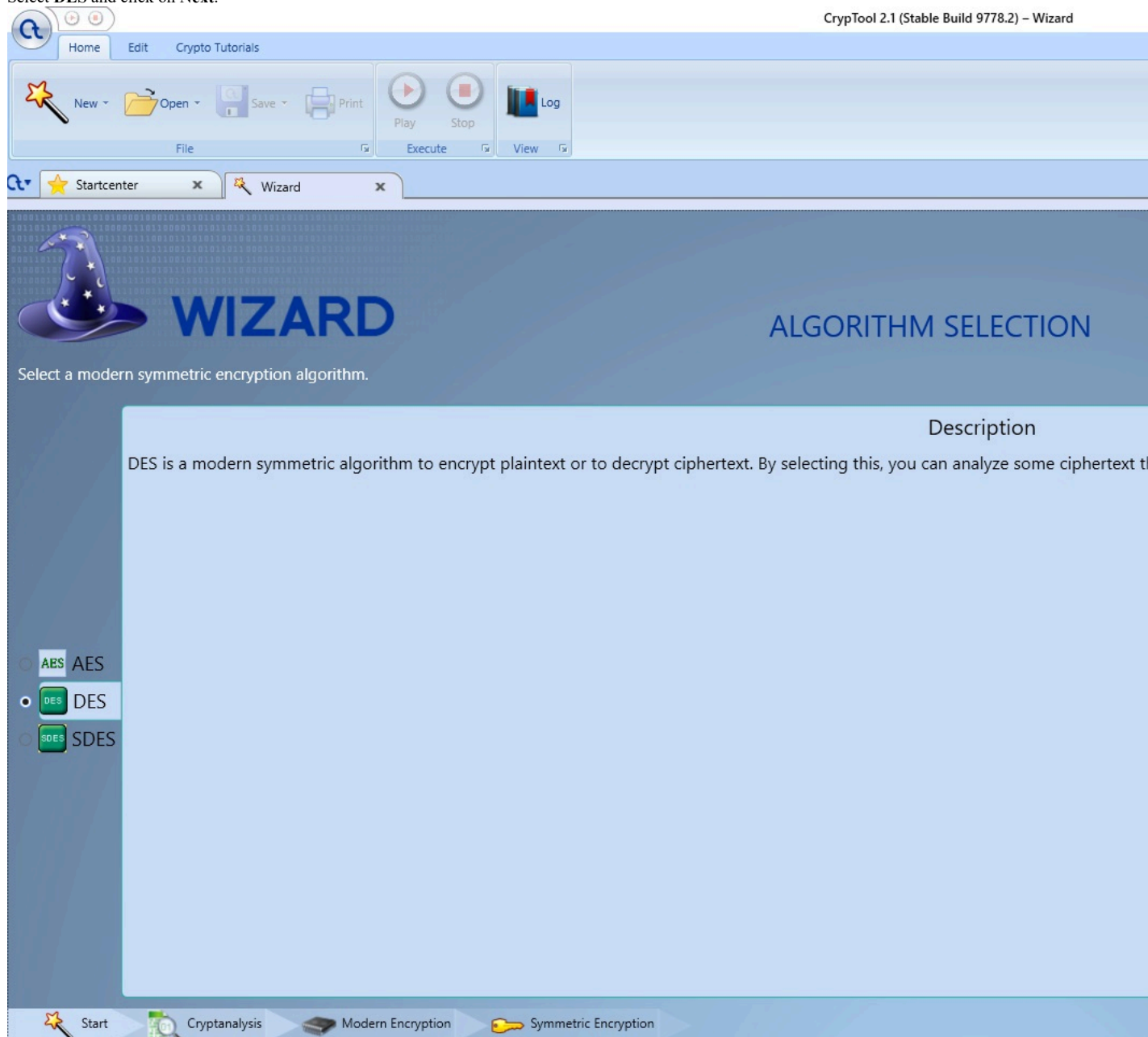
- Select **Modern Encryption** and click on **Next**.



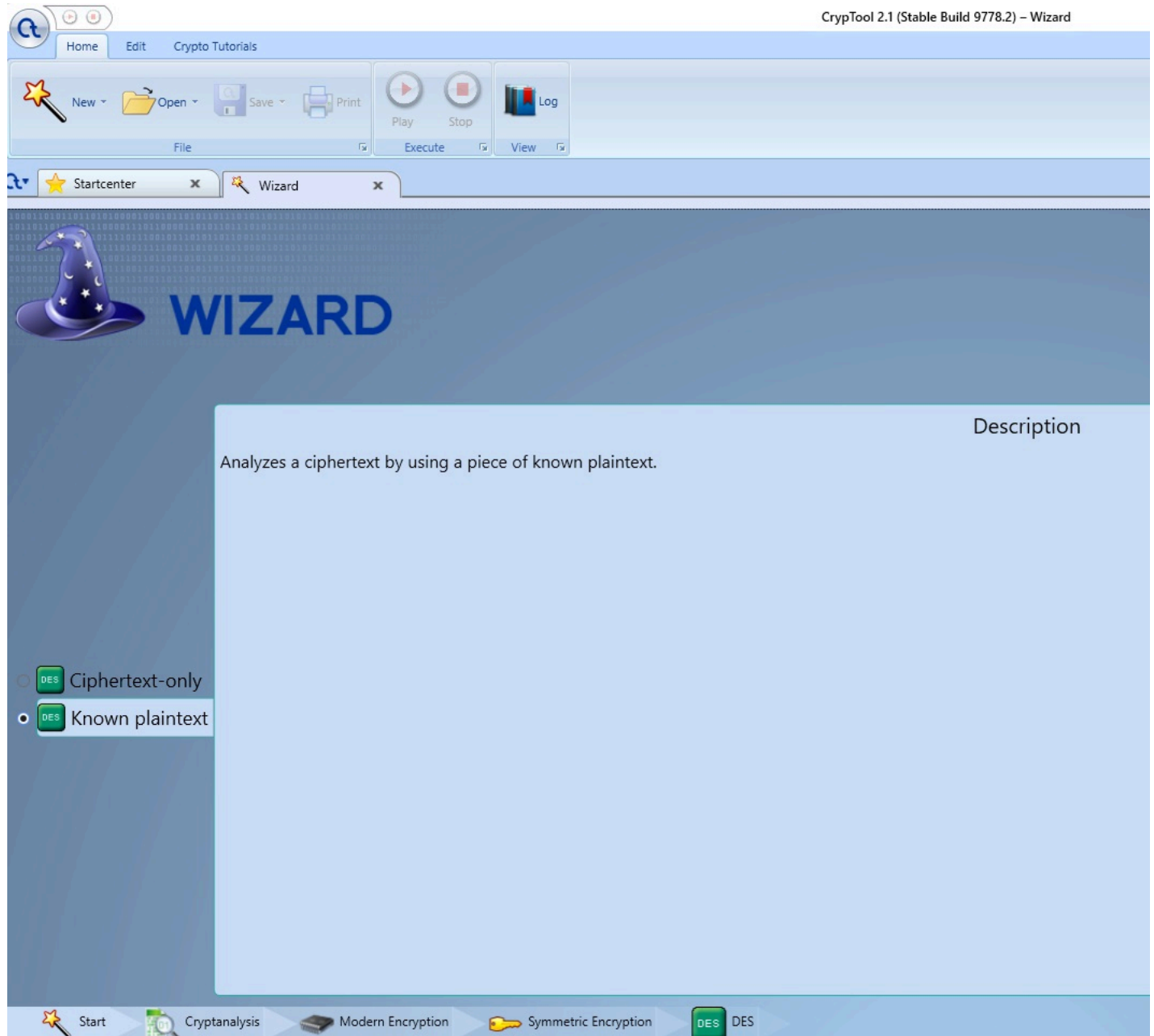
- Select **Symmetric Encryption** and click on **Next**.



- Select **DES** and click on **Next**.



- Select **Known Plaintext** and click on **Next**.



- Keep all the default values unchanged and select **Next**.

CrypTool 2.1 (Stable Build 9778.2) – Wizard

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WIZARD

DES - KNOWN-PLAINTEXT ANALYSIS

Please input the ciphertext and known plaintext.

Ciphertext to analyze (in Hex):

F75F8F1C1DA276C4C56B871BF41F473CB871E1C25436AC8518C5D43BFF8545CF

Known plaintext (as regular expression):

.*Welt.*

Keypattern:

A1-B1-C1-11-11-***-***

Chaining mode:

Electronic Code Book (ECB)

Start Cryptanalysis Modern Encryption Symmetric Encryption DES DES DES Known plaintext

- Observe the Cryptanalysis process. It will generate **Deciphered Ciphertext** and a **Key** based on analysis. The generated output may be partially or fully correct. CrypTool 2.1 (Stable Build 9778.2) – Wizard

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WIZARD

AES KNOWN-PLAINTEXT ANALYSIS OUT

Here, you can see the DES analysis.

Analyzer:

Start:9/15/2024 2:05 AMEstimated end:9/15/2024

Elapsed time:7 secondsRemaining time:

Bits to be tested:24Keys / sec:

#	Value	Key	Text
1	1	A1-B1-C1-11-11-10-10-10	Hallo Welt, das ist ein Test
2	1	A1-B1-C1-11-11-10-10-11	Hallo Welt, das ist ein Test
3	1	A1-B1-C1-11-11-10-11-10	Hallo Welt, das ist ein Test
4	1	A1-B1-C1-11-11-10-11-11	Hallo Welt, das ist ein Test
5	1	A1-B1-C1-11-11-11-10-10	Hallo Welt, das ist ein Test
6	1	A1-B1-C1-11-11-11-10-11	Hallo Welt, das ist ein Test
7	1	A1-B1-C1-11-11-11-11-10	Hallo Welt, das ist ein Test
8	1	A1-B1-C1-11-11-11-11-11	Hallo Welt, das ist ein Test
9	0	A1-B1-C1-11-11-00-00-00	~o?ücaüdBİ=Å\~W/#Epi'.ÅSM*Å
10	0	A1-B1-C1-11-11-00-00-01	~o?ücaüdBİ=Å\~W/#Epi'.ÅSM*Å

Progress:

Deciphered Ciphertext:

Hallo Welt, das ist ein Test

Key:

A1 B1 C1 11 11 10 10 10

23 characters, 1 line

Start

Cryptanalysis

Modern Encryption

Symmetric Encryption

DES DES

DES Known plaintext

DES Known-Plaintext Analysis

Exercise

- Perform the DES Cryptanalysis using "Ciphertext Only" and observe the output.

Summary

In this reading, you used CrypTool 2.1 to perform a cryptanalysis of DES encryption through a known plaintext/Ciphertext Only attack. The process involved selecting the DES algorithm and analyzing encrypted text to recover the deciphered message and key. This exercise provided insights into the weaknesses of symmetric encryption, demonstrated how known plaintext attacks work, and displayed the ability of cryptanalysis tools to decrypt messages without knowing the original key.

Author(s)

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